

This project has the capability to bring cost-effective mobility those whose motor skills are compromised.

The implementation is clever. A regular webcam, with the infrared filter removed, tracks the eye movements. The eye is illuminated by infrared light from LEDs which helps the system to operate in low-light condition. Infrared light is invisible so there is no stress on the eye. The camera is housed inside a custom-designed 3D-printed case. The video stream from the webcam is processed real-time with the help of a Raspberry Pi computer which determines the position of the pupil and according to its position converts it to values in terms of forward, backward, left and right. The move command is then verified with the help of a manual switch. Currently it's in the manual stage but eventually it will be configured to detect the tiny movements of the cheek or the tongue.

The mechanical functioning is handled by an Arduino microcontroller which controls recycled windscreen wiper motors to turn 3D-printed custom wheels which are connected to the tyres of the wheelchairs, thus moving them forward and backward. It also even features an infrared-based collision detection system.

The system is programmed using Python and OpenCV image processing library. So if you intend to replicate this project you will have to learn a bit of Python programming, which will go a long way in your other future



Image Credit: basf.de

Forget all the levers and joystick and control it with your eyes